

STRUCTURAL ENGINEERING – OCEAN TRANSPORTATION ANALYSIS FOR MODULES

09 – Derivation of Hog and Sag Deformations

This attachment provides a derivation and example calculations for hog and sag deformations using a sinusoidal curvature.

1.0 DERIVATION

Part A: determine deformation along mid-depth of ship (halfway between deck and keel) assuming symmetry about midship.

Step A1: The basic parameters are,

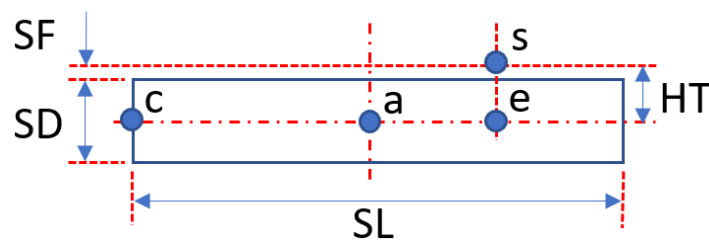


Figure 1: Undeformed Section Diagram of a Marine Vessel

Point “a” = location at mid-depth and midship

Point “c” = location at mid-depth and at fore or aft end

Point “e” = location at mid-depth and a defined horizontal distance X from midship

Point “s” = location at top of seafastening directly above point “e”

SL = ship length

SD = depth from deck to keel

SF = height of seafastening

HT = SF + SD/2

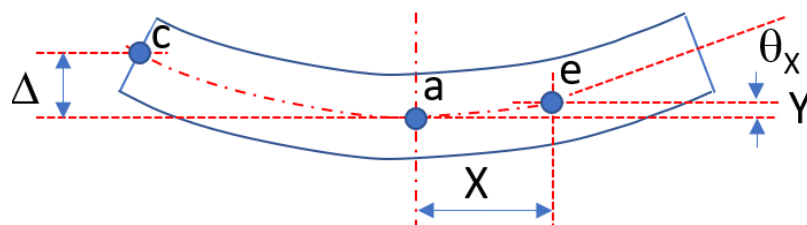


Figure 2: Sag Deformation Diagram of a Marine Vessel



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X = user-defined horizontal distance from midship to point of interest “e”

Y = vertical deformation at horizontal distance X from midship

Δ = vertical deformation due to hog/sag deformation at fore or aft end (or Y when $X = SL/2$)

θ_x = rotation angle at horizontal distance X from midship

Step A2: The basic form of a sinusoidal deformation at midship is,

$$Y = A \sin [B X + C] + D,$$

X = horizontal distance from midship (point “a”)

Y = vertical deformation from ship mid-height

A, B = scaling factors

C, D = shift factors

Step A3: Start with an initial sine function,

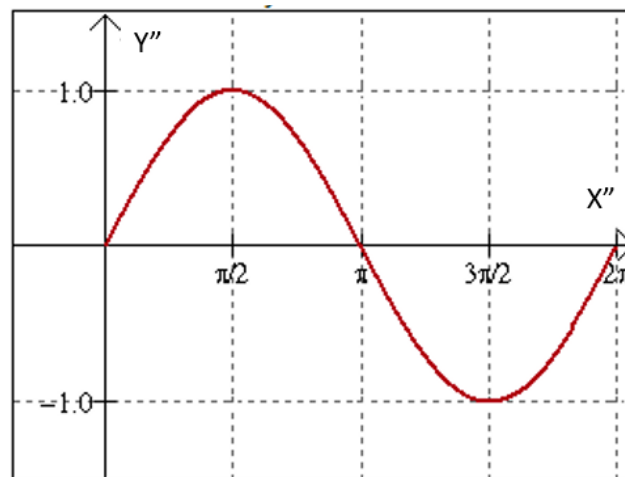


Figure 3: Initial Sine Function

In this case, $Y'' = A \sin [B X'']$, where $A = B = 1$ and $C = D = 0$, or just $Y'' = \sin[X'']$

Step A4: Rescale the sine function such that, the peak amplitude is Δ and the half period is SL .

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$$X' = X'' (2 SL / 2 \pi) \text{ and } Y' = Y'' (\Delta / 1)$$

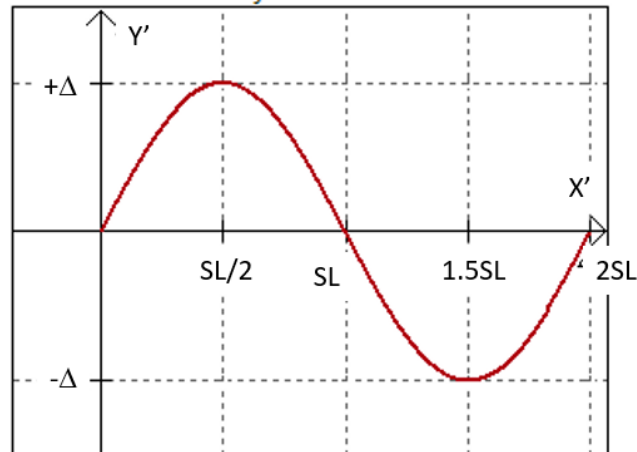


Figure 4: Rescaled Sine Function

Performing the scaling substitution results in,

$$Y'' = A \sin [B X'']$$

$$Y' (1 / \Delta) = \sin [X' \pi / SL]$$

$$Y' = (\Delta) \sin [(\pi / SL) X']$$

Thus

$$Y' = A \sin [B X' + C] + D \text{ where } A = \Delta, B = \pi / SL, \text{ and } C = D = 0$$

Step A5: Define a plot shift such that, the origin is at bottom of the sine curve.

$$X = X' - 1.5 SL \text{ and } Y = Y' + \Delta$$

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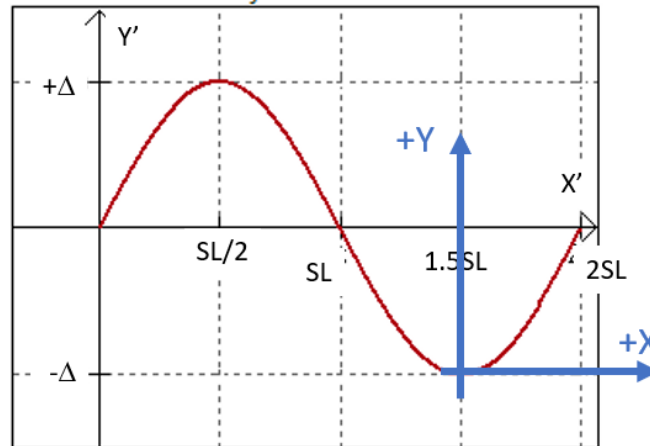


Figure 5: Shifted Sine Function

Performing the shift substitution results in,

$$Y' = \Delta \sin [\pi / SL) X']$$

$$(Y - \Delta) = \Delta \sin [(\pi / SL) (X + 1.5 SL)]$$

$$Y = \Delta \sin [(\pi / SL) X + 1.5 (\pi / SL) SL] + \Delta$$

$$Y = \Delta \sin [(\pi / SL) X + 1.5 \pi] + \Delta$$

Thus,

$$Y = A \sin [B X + C] + D \text{ where } A = \Delta, B = \pi / SL, C = 1.5 \pi, \text{ and } D = \Delta$$

Note that a more detailed derivation would indicate that the horizontal deformation at mid-depth is not zero, but it is negligible.

Step A6: Derive slope & rotation at location X

With the mid-height vertical deformation as a function of X described as,

$$Y = A \sin [B X + C] + D$$

The slope of the mid-depth deformation as a function of X is the derivative of Y,

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$$\text{slope} = A B \cos [B X + C]$$

The rotation angle θ_x of the mid-depth deformation as a function of X is,

$$\theta_x = \arctan [\text{slope}]$$

Part B: deformation at deck level

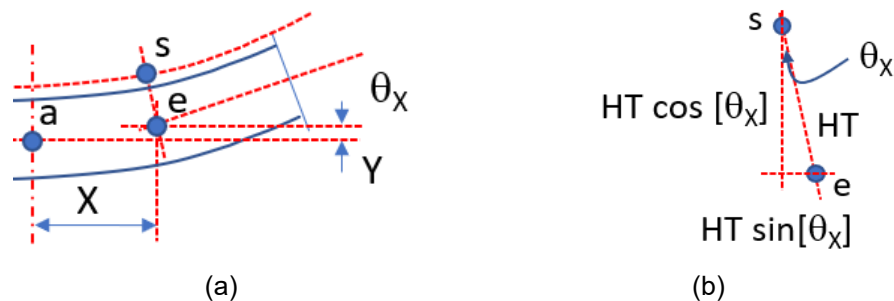


Figure 6: Enlarged Deformed Diagram

Position of point "s"	Undeformed	Deformed	Net deformation
horizontal	X	$X - HT \sin [\theta_x]$	$- HT \sin [\theta_x]$
vertical	HT	$Y + HT \cos [\theta_x]$	$Y + HT \{ 1 - \cos[\theta_x] \}$

2.0 EXAMPLE

Let ship length, SL = 160 m

Let maximum sag deformation, $\Delta = 550$ mm

Let dimension from mid-depth up to bottom of module, HT = 13 m

Calculate sag deformation at 45 m from center of module & center of deck (i.e., X = 45 m).

$$A = \Delta = 0.55 \text{ m}$$



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$$B = \pi / SL = \pi / 160 \text{ m} = 0.0196 \text{ rad/m}$$

$$C = 1.5 \pi = 4.712 \text{ rad}$$

$$D = \Delta = 0.55 \text{ m}$$

Vertical deformation at mid-depth and 45 m from center,

$$X = 45 \text{ m}$$

$$Y = A \sin [B X + C] + D = (0.55 \text{ m}) \sin [(0.0196 \text{ rad/m})(45 \text{ m}) + (4.712 \text{ rad})] + (0.55 \text{ m}) = +0.20 \text{ m, upward}$$

Calculate net deformation at deck level,

$$\text{slope} = A B \cos [B X + C] = (0.55 \text{ m})(0.0196 \text{ rad/m}) \cos [(0.0196 \text{ rad/m})(45 \text{ m}) + (4.712 \text{ rad})] = 0.0083 \text{ rad}$$

$$\theta_x = \arctan [\text{slope}] = \arctan [0.0083 \text{ rad}] = 0.0083 \text{ rad}$$

$$ND_V = Y + HT \{ 1 - \cos [\theta_x] \} = (0.20 \text{ m}) + (13 \text{ m}) \{ 1 - \cos [0.0083 \text{ rad}] \} = 0.2004 \text{ m}$$

$$ND_H = - HT \sin [\theta_x] = (13 \text{ m}) \sin [0.0083 \text{ rad}] = 0.108 \text{ m}$$